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HARMONIZATION OF ACTION PLANS AND NOISE REDUCTION PLANS: THE HUSH PROJECT

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Abstract
The project HUSH. (Harmonization of Urban noise reduction Strategies for Homogeneous action plans) has been founded by EU and is now being developed by the City of Florence, together with Vie En.Ro.Se. Ingegneria, University Florence, National and Regional Environmental Agencies, with the International Support of Eurocities Network.
In many countries a pre-existent National legislation about noise mapping and noise action planning has been used and cannot be simply replaced by the END Directive. In Italy there are now three legislation disposals concerning environmental noise action planning. All the former EU granted projects (like Harmonoise, Imagine, Silence and others), have not considered this aspect of coexistence of plans and all the problems generated by it.
Moving from this, the project is oriented towards the harmonization of the different Plans provided by the different legislations and aims to provide a general methodology.
Taking in account the significant experiences of action planning in European Member States as well as some general recommendations drawn by Working Groups established by European Commission to support the END Directive, the project will lead city administrators and policy makers to the integrated action planning approach: only one plan instead of two or more, with consequent reduction of unnecessary duplicate action and of their relative costs.
In this paper the general setting and the startup actions of the project are shown.

Keywords: harmonization, actions, plan, integration, project

1 Introduction

In 2008 Comune di Firenze has set up a project which focuses on the harmonisation of urban noise reduction strategies. The city has applied for an EU grant under the LIFE+ programme and the project proposal was approved. Since the beginning of 2010 all the beneficiary partners (the authors of this paper represent two of them) are workin on the development of the respective actions.

2 EU cities scenarios and targeted problem

2.1 Noise and city planning

According to the June 2007 Survey of European Commission DG Regional Policy, almost 3/4 of European citizens are living in urban areas today, and urban dwellers are expected to increase to represent 80% of the total population by 2020. In many respects, the European Union can be seen as a Union of cities, about 1600 settlements with over 50.000 inhabitants are considered as functional urban areas. So cities are the places where environmental challenges are focused and where quality of life is determined according to socio-economic and political factors. Quality of life and quality of the environment fundamentally reflect how well towns and cities are functioning or do not. Towns and cities are also the places where the challenges of sustainable urban development must be addressed.
Noise in large urban agglomerations is now seen as a factor that greatly impairs quality of life, similarly to air or water pollution. In accordance to the European Commission's Green Paper on future noise Policy, more than about 250 million people are exposed to A-weighted outdoor levels higher than 55 dB. The European Directive 2002/49/EC on the Assessment and Management of Environmental Noise (END) was adopted to define a common approach to avoid, prevent or reduce the harmful effects due to exposure to noise. To that end, the EU Commission required the Member States to produce noise maps for the main sources of noise pollution (road traffic, railways, airplanes, factories), described the indicators to be used (namely Lden and Lnight) and, in later documents, suggested the methodologies to be followed and the algorithms for modelling noise emissions from the different sources. The work of the EU commission has been integrated by the research produced by WG-AEN, WG-HSEA and in the context of relevant EU-funded projects, which have pointed out the best practices to be shared among the Community.
In quite all European cities annoyance is cause of discomfort and strongly affects the perception of quality of life. It is part of European Community policy to achieve a high level of health and environmental protection, and one of the objectives to be pursued is protection against noise. Reduction of noise in terms of reduction of exposure to harmful environmental noise in urban areas is the most important expected result.
A strategic plan is a description of a set of activities with their expected or intended time planning. This means that an action plan should define actions in a ‘smart’ way (specific, measurable, achievable, realistic, time related). So, an action plan should not merely be a policy statement. It should describe the targets to be achieved, i.e. the effects in terms of noise exposure expected from the concrete actions to be taken in the period until the next noise map. A good noise action plan is accountable, i.e. it allows both the politician and the citizen to compare expected and real effects and to revise the policy if necessary.

2.2 END Directive and National Regulations

This project refers to one of the main problems that affect quality of life in urban areas: noise and its effects on population.
The 2002/49/CE Directive, known as Environmental Noise Directive (END). In the Directive "environmental noise" shall mean unwanted or harmful outdoor sound created by human activities, including noise emitted by means of transport, road traffic, rail traffic, air traffic, and from sites of industrial activity. Member States shall produce strategic noise maps, and strategic action plans. The Directive ask the Member States to provide:
- strategic noise maps, finalized to establish global exposure to noise in a certain area, due to different sources working together, or to produce general estimations to that area.
- action plans, aiming to solve acoustic pollution problems and related effects, including, if necessary, its reduction.
Action Plans are designed to manage, within their territories, noise issues and effects, including noise reduction if necessary, and shall aim to protect quiet areas against an increase in noise. The measures within the plans should notably address priorities which may be identified by the exceeding of any relevant limit value or by other criteria chosen by the Member States, which must ensure that, no later than 18 July 2013, the competent authorities have drawn up action plans addressing priorities. The action plans shall be reviewed, and revised if necessary, when a major development occurs affecting the existing noise situation, and at least every five years after the date of their approval.
In many countries a pre-existent National legislation about noise mapping and noise action planning has been used and cannot be simply replaced by the Directive.
In Italy there are now three legislation disposals concerning environmental noise action planning: L. 447/1995 requires to City Councils the Local Action Plans derived by Acoustical Classification of territory; D.M. 29-11-2000 requires to Transportation and Infrastructures companies their Noise Reduction Plans; finally D.Lgs. 194/2005 put in the National legislation all the requirements of the END directive. Moving from these last point, the project is oriented towards the harmonization of the different Plans provided by the different legislations.

2.3 Former Projects and HUSH

Some projects have been developed since 2002, all their results must be taken in account, but no one of them seem to be addressed to solve the over-planning problem mentioned above.
HARMONOISE project, has produced methods for the prediction of environmental noise levels caused by road and railway traffic. These methods are intended to become the harmonized methods for noise mapping in all EU Member States. The methods are developed to predict noise levels in terms of Lden and Lnight, which are the harmonized noise indicators according to the Environmental Noise Directive 2002/49/EC.
IMAGINE project, has developed calculation methods for railway, road, industrial and aircraft noise, standardising the Harmonoise methods and providing guidelines on how to use these methods for noise mapping and noise action plans (e.g. traffic flow management) in the EC.
CALM II project has provided enhanced coordination of the European transport noise research in all related research areas.
QCITY project is an integrated research project for the development of an integrated technology infrastructure for the efficient control of road and rail ambient noise by considering the attenuation of noise generation at source at both vehicle/infrastructure levels.

SILENCE project has developed an integrated methodology for the improved control of surface transport noise in urban areas.
Other important useful references for the state of art are those deriving from the European Network EUROCITIES and, in particular, from the Working Group Noise.
Finally all the significant experiences of action planning in European Member States as well as the some general recommendations drawn by Working Groups established by European Commission to support the END, have to be taken in account.
But there are remarkable aspects of this project, not considered in such a way from any of the projects listed above, listed below.
a) integrated action planning: only one plan instead of two or more, with consequent reduction of unnecessary duplicate action and of their relative costs;
b) integrated solutions' designing methods taking into account a single-value priority index that combines those given by different plans;
c) integrated interventions for areas with concurrent sources.
Particular attention will be put on the planning of actions (and on designing of relative solutions) regarding the protection of quiet areas, not only with respect to acoustic criteria, but also to such items as future land use etc.
In these areas, a wide range of requirements must be met at the same time, as different users could require and expect a different level of "quietness". The expectation of the people involved, then, appears to be of primary importance: new indicators have to be considered, among them the soundscapes descriptors and other index of people's noise perception.

3 The project HUSH

3.1 Main and specific objectives of HUSH

Harmonization of Urban noise reduction Strategies for Homogeneous action plans (HUSH) is the name and the program of this project: it moves from the evidence that harmonization of noise action planning methods is needed, not only in Italy but also in all the European countries where a former Legislation about noise planning was present at the moment of END Directive adoption.
The general objective is harmonizing national noise management standards with European Directive 49/2002 to obtain homogeneous noise Action Plans, contributing to the more general need of transposing, implementing and enforcing a common or harmonized environmental legislation between EU countries.
Specific Objectives of the project are:
1. to point out unsolved conflicts among current standards at Regional, National and European level, and to define common methods for designing strategic and specific solutions;
2. to define a new development sistem (procedures and database) for action planning by testing it in a pilot case;
3. to design guidelines in order to build a system for action plan applications, to support Regional, National and European Law reviews.

3.2 Synthetic agenda of HUSH project

Actions and timing of the project are shown in the milestones list of table 1 and in the timetable of figure 1. The contents and the means of the various actions are summarized in the following paragraph.

Table 1 – Milestones of HUSH and relative timing

Name of the Milestone	Code of the associated action	Deadline
General data collection	1	30/03/2010
Data collection to design work platform	2	30/03/2010
Detection and classification of normative conflicts between National and European laws	3	30/03/2010
Detection and classification of normative conflicts between Regional and National laws	4	30/03/2010
Data collection of noise reduction measures	5	30/03/2010
Detection of methodological conflict solutions	6	30/06/2010
Website construction and test	19	30/06/2010
System (database and procedure) definition	7	30/09/2010
Criteria definition to choice pilot areas and procedures for system testing	7	30/09/2010
Analysis of strategic interventions	11	30/09/2010
Ante-operam noise measurement campaign	8	31/12/2010
Ante-operam non-acoustic data acquisition	10	31/12/2010
Acoustic and civil design of direct interventions	9	30/03/2011
Execution of direct interventions	12	31/12/2011
In progress control of intervention execution	13	31/12/2011
Post-operam non-acoustic data acquisition	14	30/04/2012
Post-operam noise measurement campaign	15	30/04/2012
System (database and procedure) optimization	16	30/05/2012
Proposal for revision of both Italian Legislation and END Directive	17	30/06/2012
Proposal for revision of Regional Legislation	18	30/06/2012
Dissemination	19	31/12/2012
Monitoring	20	31/12/2012

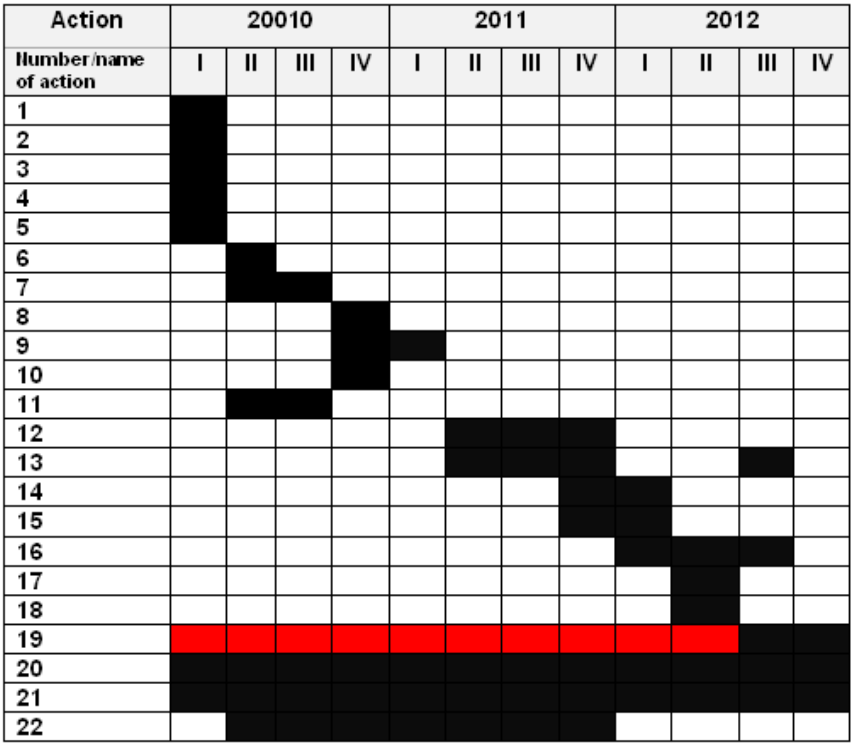


Figure 1 – Timetable of HUSH scheduled actions

3.3 Summary of HUSH project architecture

The plan of development of HUSH activities is summarized in the following list of actions
Action 1 – Florence Municipality shall collect and organize all the general data already available in both existing study platform for Florence Action Plan and local noise abatement plans. At the meantime, the national and local transport companies shall provide all data concerning the planned actions for reduction of transport system noise. Secondly, all these databases will be compared and put together in the same Geographical Information System (GIS).
Action 2 - This action is aimed to analyse and compare available (in Italy and elsewhere) data platforms needed to address the requirements set out by National, Regional and European regulations. The comparison of databases and methods available in Italy and in cities of other European countries will contribute to define the H.U.S.H. work platform.
Action 3 - Activities will focus on detection and collection of National and European Acoustics laws and standards published for different contexts and timing and concerning noise reduction action planning as well as management of noise. Given the general and specific objectives of harmonization of the H.U.S.H. project, a specific indicator will be defined to express the quantity of Normative Conflicts between National and European levels (NCNE). This index will be used, together with others, to monitor and evaluate the overall impact of H.U.S.H.
Action 4 - Activities will focus on detection and collection of Regional Acoustics laws and standards published for different contexts and timing and concerning noise reduction action planning as well as management of noise. Normative conflicts between regional and national levels will be pointed out. Given the general and specific objectives of harmonization of the

H.U.S.H. project, a specific indicator will be defined to express the quantity of Normative Conflicts between Regional and National levels (NCRN). This index will be used, together with others, to monitor and evaluate the overall impact of H.U.S.H.
Action 5 - Collection of solution cases for the reduction of noise in urban areas. Such collection will provide a significant contribution to further development of the work platform. Collected data will be analysed by qualified engineers and compared with the state of art. Starting from both data analysis and collected experiences, problems encountered during the acoustic design phase will be catalogued and solved.
Action 6 – Given the harmonization aims of H.U.S.H., all the critical arguments identified in Actions 3 and 4 will be grouped by similarities and methodological solutions will be defined.
Action 7 – This action will verify feasibility of proposed methodological solutions pointed out by the ISPRA activity in Action 6. The Geographical Information System (GIS) will be built up. Procedures and data will be implemented into the GIS environment, according to goals and solutions proposed in previous actions. Pilot cases will be selected and data acquisition criteria will be defined in order to validate the whole system.
Action 8 – The planning of interventions and the selection of areas in which interventions are to be realized are acquired. In each selected area specific noise measurement campaigns (long and short period measures) will be carried out.
Action 9 – The Acoustic and civil design of strategic interventions (road systems, traffic regulations, transports) and of direct interventions on pilot areas will be performed.
Action 10 – The planning of interventions and the selection of areas in which interventions are to be realized are acquired. In each selected area specific non-acoustic data will be carried out in Action 10 and 14, that are strongly linked. Action 10, ante-operam, will focus on collection of all data used in the Florence Action Plan draft. Among those, notions about receivers, with particular attention to sensitive ones (schools and hospitals), and inquiry on people's disturb perception. Action 14, post-operam, will focus on an inquiry on people's contentment about implemented abatement solutions (e.g. Noise barriers).
Action 11 – Typological experimentation of strategic interventions for urban traffic, transportation systems and mobility plans will be defined.
Action 12 – Planned interventions will be put into execution, according to acoustical and civil design produced in Action 9. For each solution a specific task will be defined and will become object of specific entrusting to a specialized firm.
Action 13 – In progress control execution of pilot cases' interventions and possible changes to the plan.
Action 14 – Acquisition of non-acoustical post-operam data in pilot cases. Acquisition of integrating data for procedure optimization.
Action 15 – After each intervention, a detailed noise measurement campaign will be carried out. For each intervention, noise measurements will be done, under controlled traffic conditions, in the key test positions.
Action 16 – This action aims to guarantee both the system optimization and evaluation. In particular, database and procedures will be optimized referring to experimental data obtained by pilot cases. System evaluation will be carried out by using indicators defined in previous actions
Action 17 – Proposals for revision of National Legislation and END Directive (in the frame of periodic review of legislation). Spread of results among the partners, issuing of a report for the Ministry and diffusion within the Italian Environmental Agencies System
Action 18 – Proposals for revision of Regional Legislation. Spread of results among the partners and issuing of a report for the Tuscany Region and diffusion within the Regional Environmental Agencies System.
Action 19 – Final Report of Activities. Dissemination through dedicated website and printed matters.

Action 20 – The objectives of this action are: to monitor the compliance of carried out activities with those foreseen in the grant agreement (project proposal) and to evaluate the impact of the project. Furthermore, an analysis of the cross cutting issues will be carried out including: Management and coordination; Project sustainability; Dissemination plan post-project-end; General Impact on the carbon footprint.

4 Expected results

The implementation of HUSH project will lead to three main categories of output:
- harmonized procedures for the development of Integrated Strategic Action Plans, considering the combined requirements of END Action Plans, Noise City Plans and Transport Noise Reduction Plans;
- harmonized procedures for the designing of the planned solutions;
- guidelines in order to build a system for action plan applications, to support Regional, National and European Law reviewers;

In the project agenda there are actions in which all the project output items listed above will be validated by means of their application to pilot cases, each one of them chosen as representative of a universe of homogeneous cases.
All the existing different methods of action planning for noise reduction and/or noise control will be investigated and described and a matrix of kowledge referred to European cities noise plans will be considered. As a final result, a summary of adopted solutions is presented.

5 Conclusions

The problem of matching END Directive requirements with National regulations, standards and methods is one of the main critical point to be removed in END revision and, before, to be solved by National authorities, hopefully in a harmonized way.
The harmonization of Plans can lead to important savings in terms of costs for design and realization of designed solutions. Considering the high level of optimization that can be reached applying the methodology to large scale planning, an approximate quantification of the achieved reduction of costs can be found around one fourth of the cost of all interventions for a city agglomeration.
The project could finally generate an international board, where acousticians and researchers, together with planners, architects, engineers, will meet their own experiences in design, specification, and construction of interventions, providing new international standards and compatibility rules.

References

[1] Directive 2002/49/EC of the European Parliament relating to the assessment and management of environmental noise. Official Journal of the European Communities L 189/12, 2002

[2] Working Document Working Group Noise EUROCITIES “Evaluation of the Environmental Noise Directive”, 2008

[3] Night noise guidance WHO, 2008

[4] “Green Paper on Future Noise, Towards a new culture for Urban Mobility”, COM 2007/55, European Commission, 2007

[5] Noise Questionnaire EUROCITIES 2008

[6] Florence City Council Strategic Action Plan, 2009

[7] H. Wolfert “Noise in cities: general approach and European network solutions” in Proceedings of “European strategies for noise reduction and management”, 2009

[8] S. L:uzzi “Strategies for integration of noise plans” in Proceedings of “European strategies for noise reduction and management”, 2009

[9] Florence City Council Strategic Action Plan, 2009

[10] R. Bellomini et al. “Dalle mappature acustiche strategiche ai piani d’azione: Aspetti problematici e soluzioni” in Proceedings of 36° Congress Italian Society of Acoustics (AIA), 2009

[11] H. Wolfert “Cities cannot solve their noise problems solitairly” in Proceedings of “NAG-DAGA Congress”, 2009

[12] S. Luzzi, R. Bellomini “Harmonised Strategies for the Development of Noise Action Plans”, in Proceedings of “NAG-DAGA Congress”, 2009

[13] www. comune.firenze.it

[14] www.vienrose.it

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